

Chapter 24

MIRRORS, LENSES, AND IMAGING SYSTEM

In this chapter we will study the following:

- Properties of image formed by plane and spherical mirrors (Plane, convex and Concave Mirrors)
- Properties of image formed by spherical lenses (Convex and Concave Lenses)
- Lensmaker's equation
- Power of the lens
- Solve Examples

Chapter 24

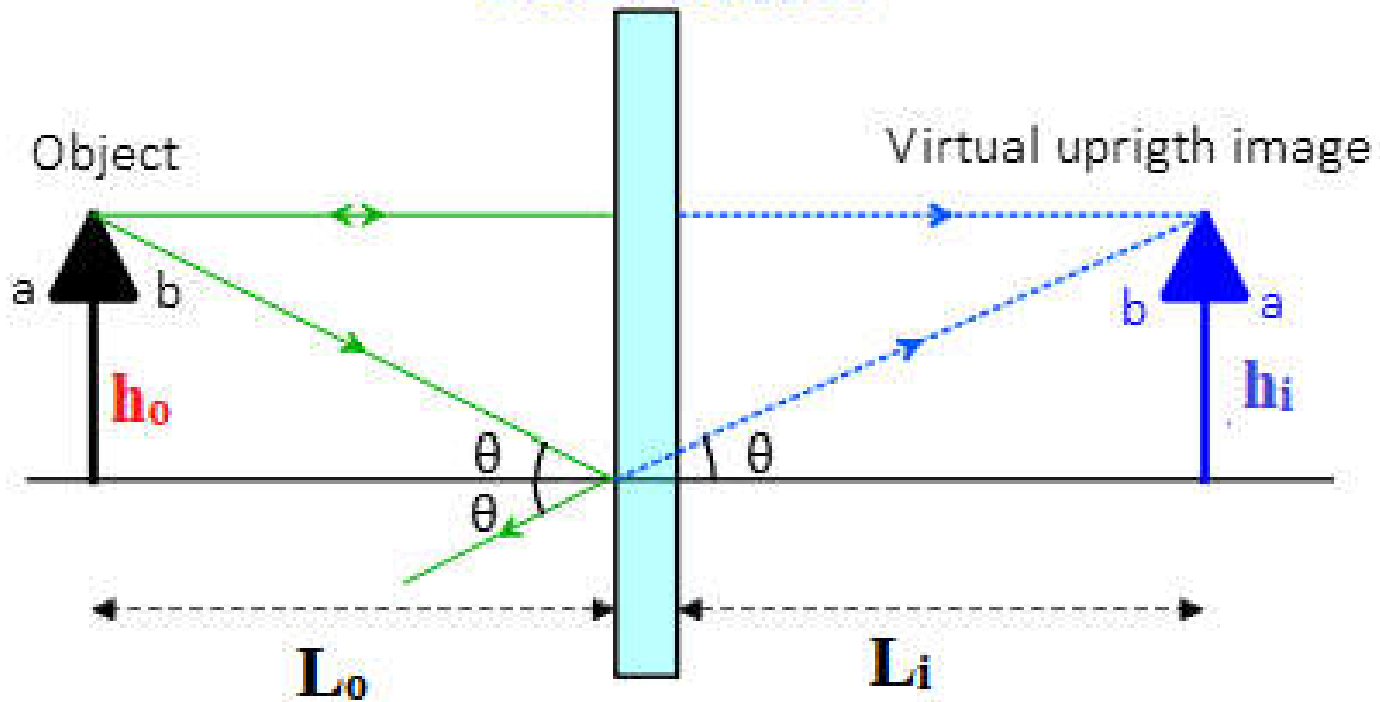
MIRRORS, LENSES, AND IMAGING SYSTEM

24.1 Mirrors

Properties of an image formed by the plane mirror:

- The image obtained is virtual.
- The image is laterally inverted.
- The image is erect.
- The size of the image is the same as the size of the object.
- The distance between the image obtained is the same as the distance between the object from the mirror.

Plane Mirror

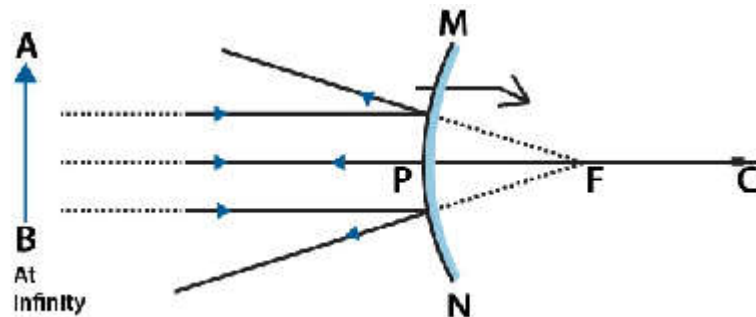


$$L_o = L_i$$

$$h_o = h_i$$

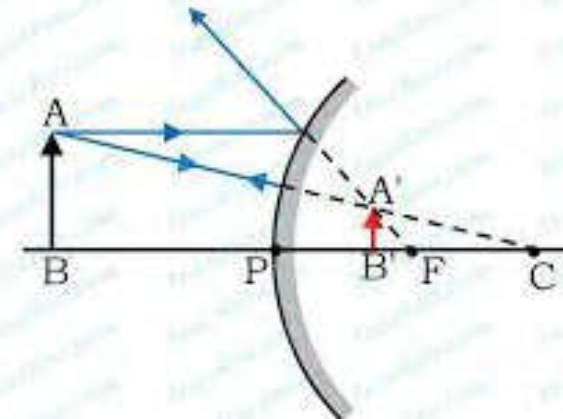
Properties of Image Formed by a Convex Mirror:

- image is always virtual.
- The image is always erect.
- The image is always diminished in size (i.e., smaller than the object)
- It is always formed between Focus and Pole.
- The object and image are on either side of the mirror.



Convex Mirror -

Object at between Infinity and Pole

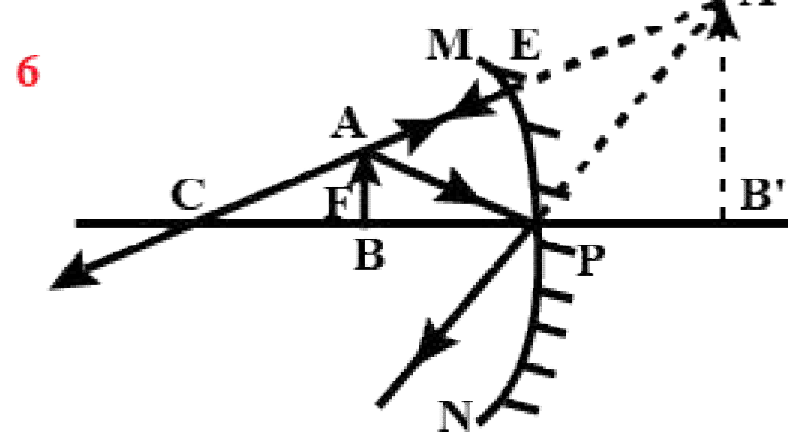
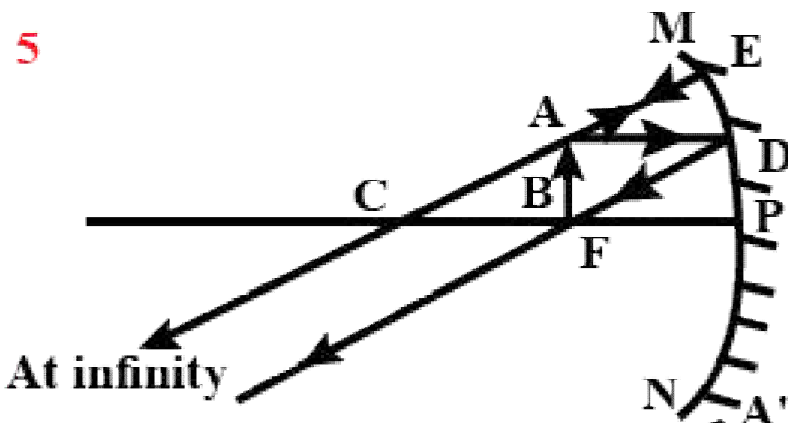
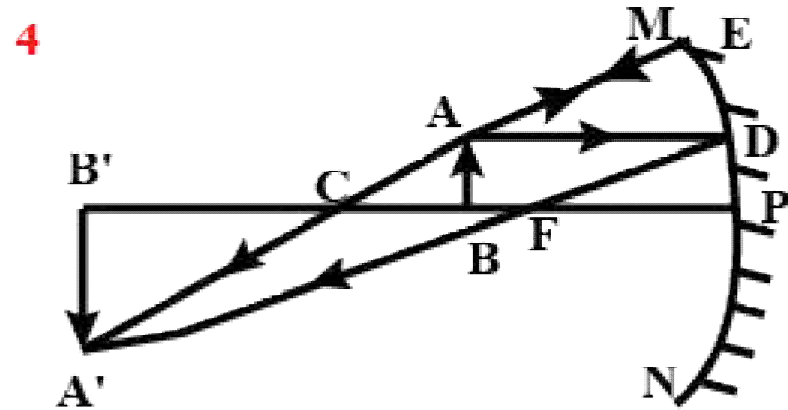
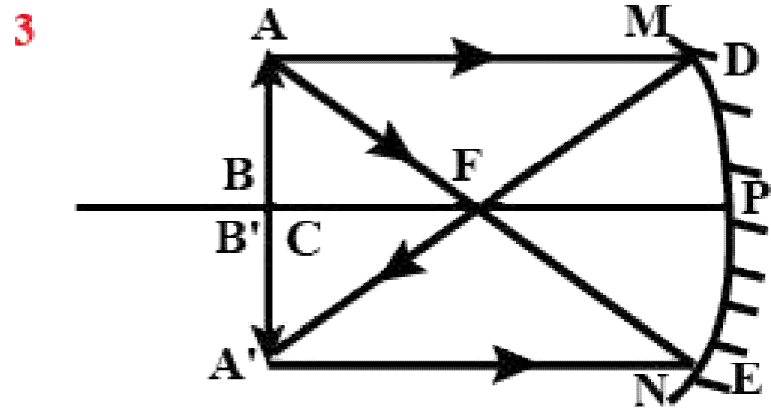
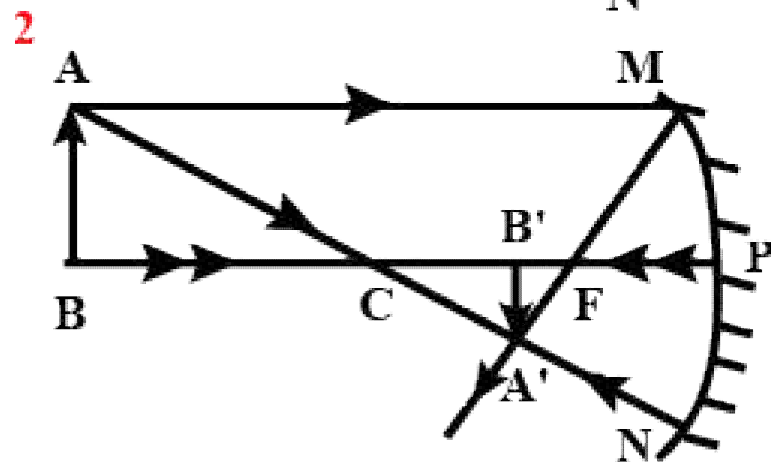
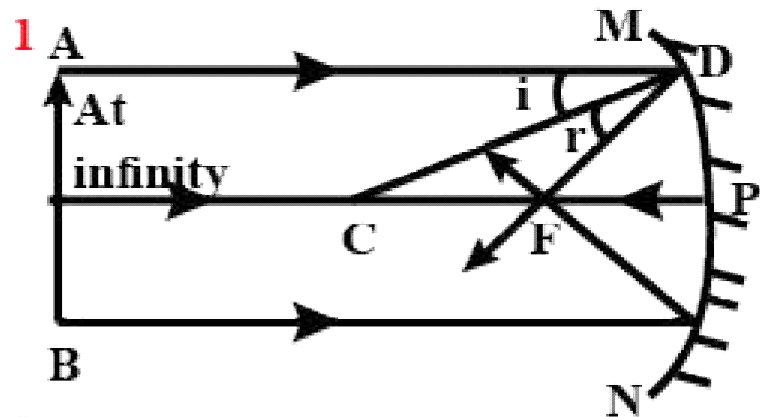


Properties of Image Formed by a Concave Mirror

Concave mirrors can produce both real and virtual images; they can be upright (if virtual) or inverted (if real); they can be behind the mirror (if virtual) or in front of the mirror (if real); they can also be enlarged, reduced, or the same size as object.

f = focal length.

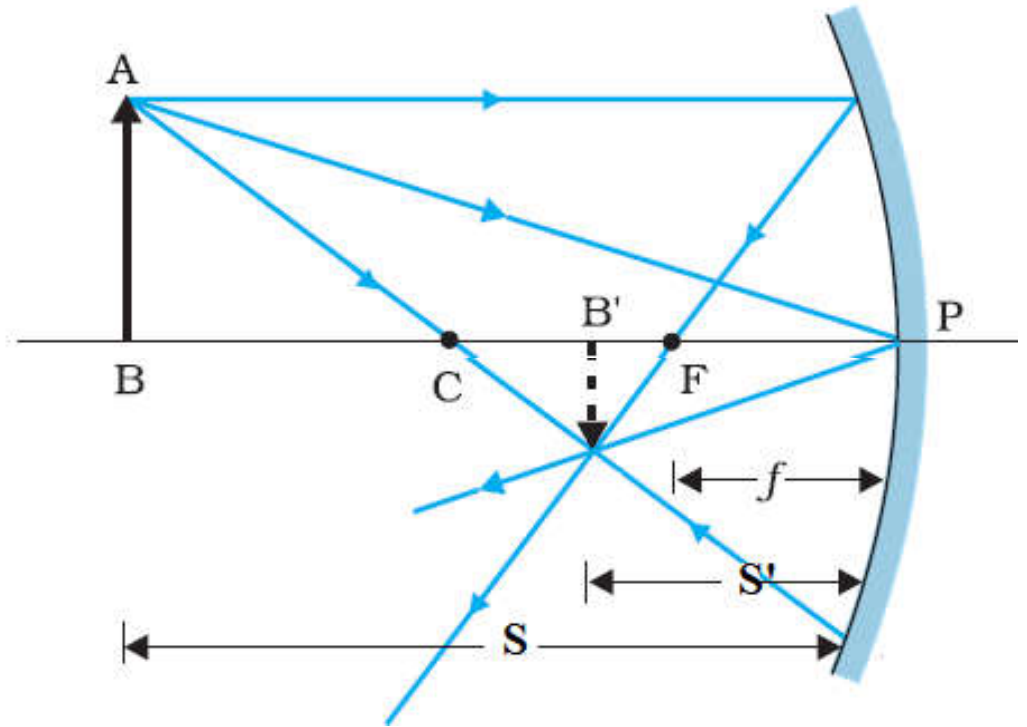
R = radius of curvature = $2f$ (or R is the radius of the sphere)



$$R = C = 2f$$

The relation that holds for the spherical mirror is given by: $\frac{1}{f} = \frac{1}{s} + \frac{1}{s'}$ This law holds for convex and concave mirrors

f = focal length
 s = the distance of the object from the mirror
 s' = the distance from the image to the mirror

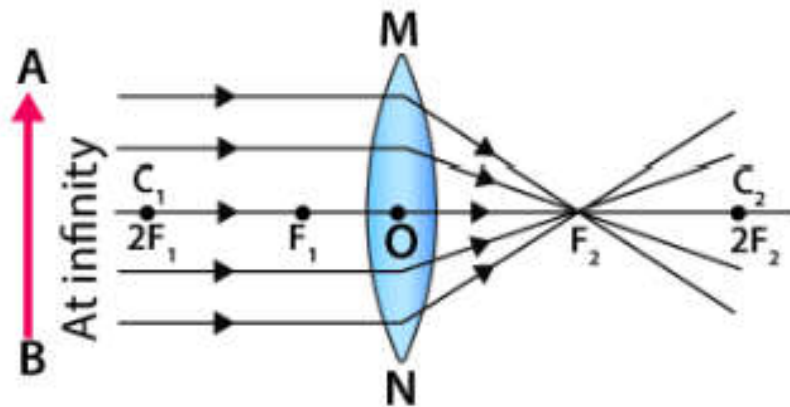


f is +ve for concave mirror and -ve for convex mirror

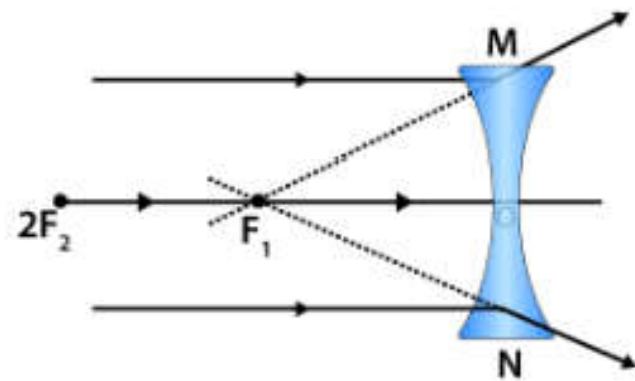
LENSES:

Two types of lenses :

- Convex lens
- Concave lens

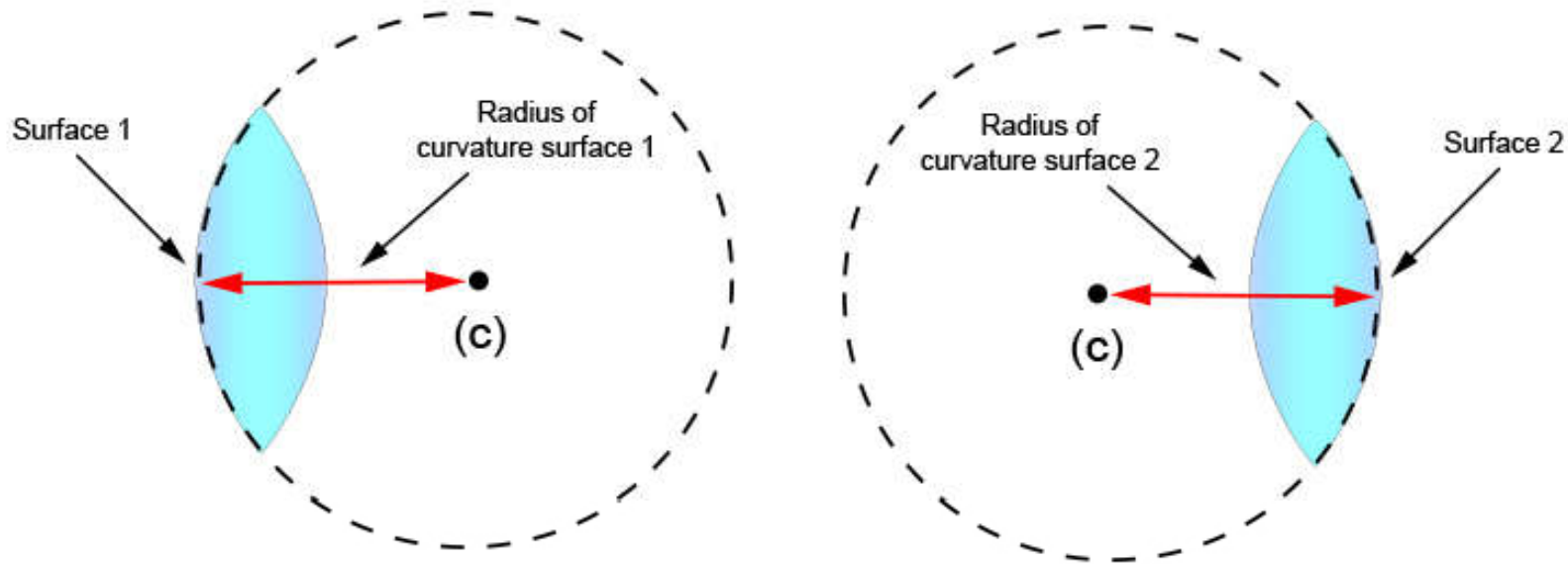


Convex Lens



Concave Lens

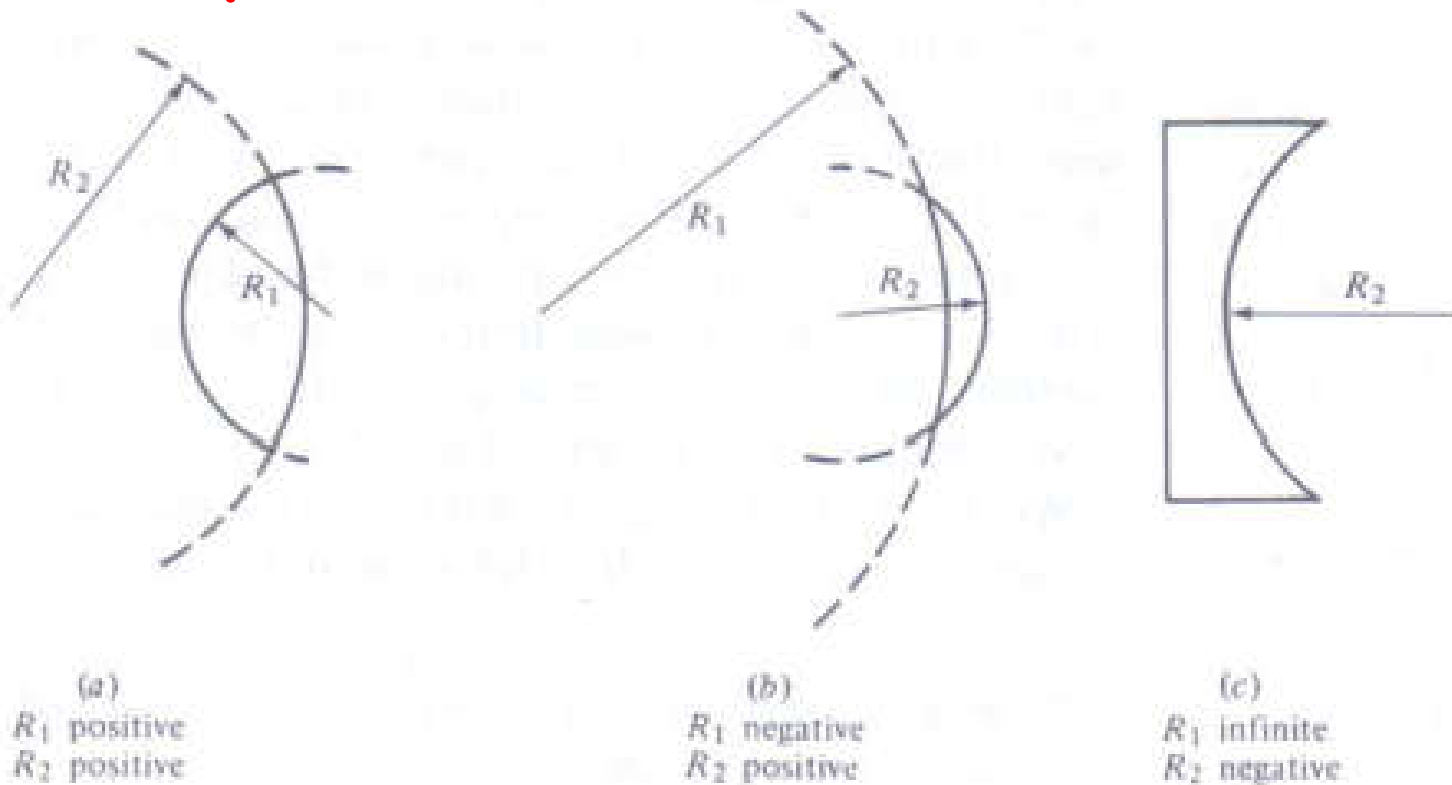
Radius of Curvature:



For a convex lens, the radius of curvature is positive while for a concave lens the radius of curvature is always negative.

Conventions for radii of curvature:

The surfaces (a) convex, convex (b) concave, convex (c) plane, concave



- Convex surface has **+ve** radius of curvature
- Concave surface has **-ve** radius of curvature
- A plane surface has **infinite** radius of curvature

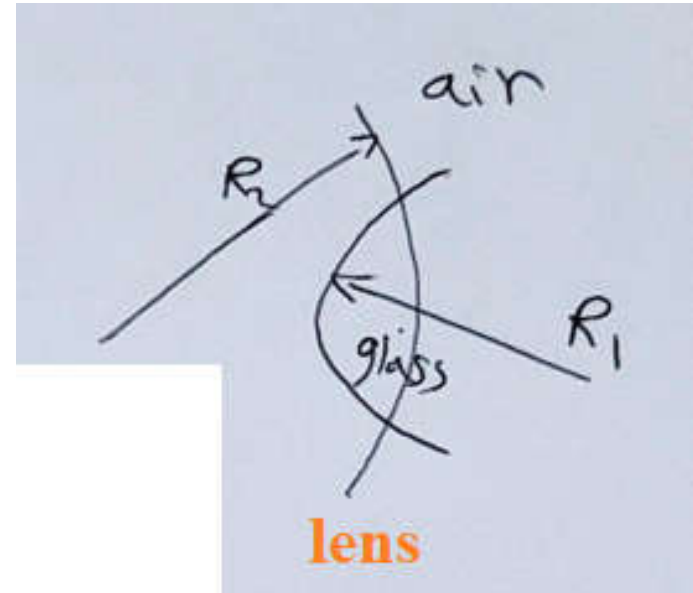
Lensmaker's Equation:

$$\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

$$n = \frac{n_{lens}}{n_{medium}}$$

Example:

In the lens shown $R_1 = 0.1\text{m}$
 $R_2 = 0.2\text{m}$. What is f ?



Solution:

$$n_{\text{air}} = 1, n_{\text{glass}} = 1.5$$

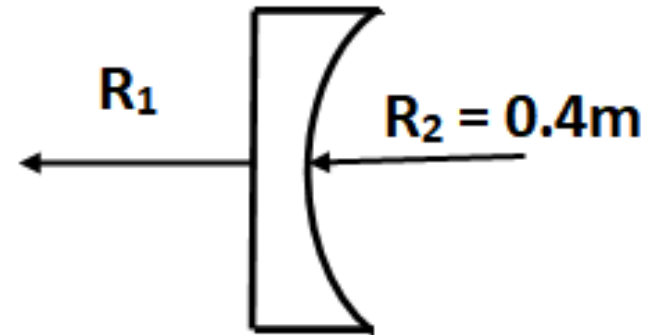
$$\frac{1}{f} = \left(\frac{1.5}{1} - 1 \right) \left(\frac{1}{+0.1} + \frac{1}{+0.2} \right)$$

$$f = +0.133\text{m}$$

NOTE: According to the Figure both R_1 and R_2 are positive

Example:

In the lens shown $R_1 = \infty$
 $R_2 = 0.4\text{m}$. What is f ?



Solution:

$$\frac{1}{f} = \left(\frac{1.5}{1} - 1 \right) \left(\frac{1}{+\infty} + \frac{1}{-0.4} \right)$$

$f = -8\text{m}$ (-ve sign indicates diverging lens)

NOTE: R_1 is ∞ because the surface is flat
 R_2 is -ve because the surface is concave

Solving Problems:

Example 24.3: A lens has a focal length of +0.1m.
Find the image distance when the object is
(a) 0.5m, (b) 0.08m

Solution

$$(a) \quad \frac{1}{s'} = \frac{1}{f} - \frac{1}{s} = \frac{1}{+0.1} - \frac{1}{0.5} = 8\text{m}^{-1}$$

$S' = 0.125\text{m}$ S' positive means real image

$$(b) \quad \frac{1}{s'} = \frac{1}{f} - \frac{1}{s} = \frac{1}{+0.1} - \frac{1}{0.08} = -2.5\text{m}^{-1}$$

$S' = -0.4\text{m}$ S' is negative means
the image is virtual

Magnification: $m = -s'/s$

In the previous example part (b)

$$m = -s'/s = -(-0.4)/0.1 = 4$$

NOTE: in part (a) the size of image is smaller than the size of the object. **Why?**

The power of the lens; Aberrations:

The power of the lens P is defines as:

$$P = 1/f$$

P is measured in diopters

$$1 \text{ diopter} = 1 \text{ m}^{-1}$$

For two lens with focal lengths f_1 and f_2 placed next to each other are equivalent to a single lens of focal length f given by: $1/f = 1/f_1 + 1/f_2$

Hence, the power of the pair lens P is:

$$P = P_1 + P_2 \quad (P_1 = 1/f_1, P_2 = 1/f_2)$$

Example:

Find the focal length of lens with a power of 4 diopters

Solution:

$$P = 1/f$$

$$f = 1/P = \frac{1}{4} = 0.25 \text{ m}$$